GradientCIO

PORTFOLIO ATTRIBUTION COMPARISON

Northstar Total Fund Attribution Comparison

Historical and forward attribution · available evidence review

PORTFOLIO
Northstar Total Fund

BENCHMARK
Policy benchmark

COHORT
Total fund

AVAILABLE VIEWS
Historical and forward

OVERALL VIEW
Two governed lanes available

SCOPE REVIEWED
2 of 4 · Partial

01 Executive Comparison

Two governed lanes are available without server comparison metrics

OVERALL VIEW

Two governed lanes available

SCOPE REVIEWED · 2/4

BOTTOM LINE

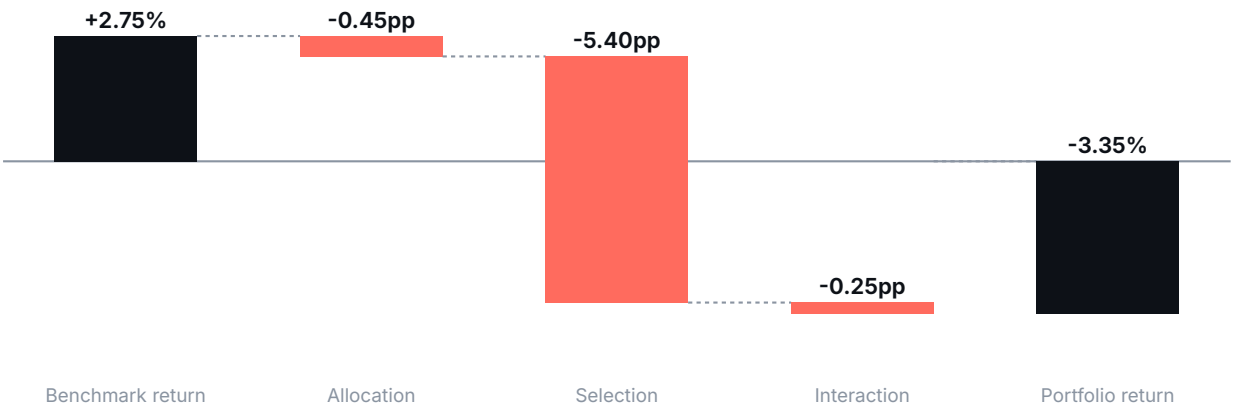
Realized shortfall comes from selection; expected gain comes from selection; Alternatives drives both S2 S3.

HISTORICAL LANE	EX ANTE LANE	SIMULATED LANE	COMPARISON COVERAGE
Available	**Available**	**Unavailable**	**Fallback**
Realized 12 months · 2025-10-31 to 2026-09-30 S2	1-year expected S3	Not available — planned per get_gradient_capabilities S1	Not available — planned per get_gradient_capabilities S1

Realized shortfall from selection; expected gain from selection; Alternatives drives both S2 S3.

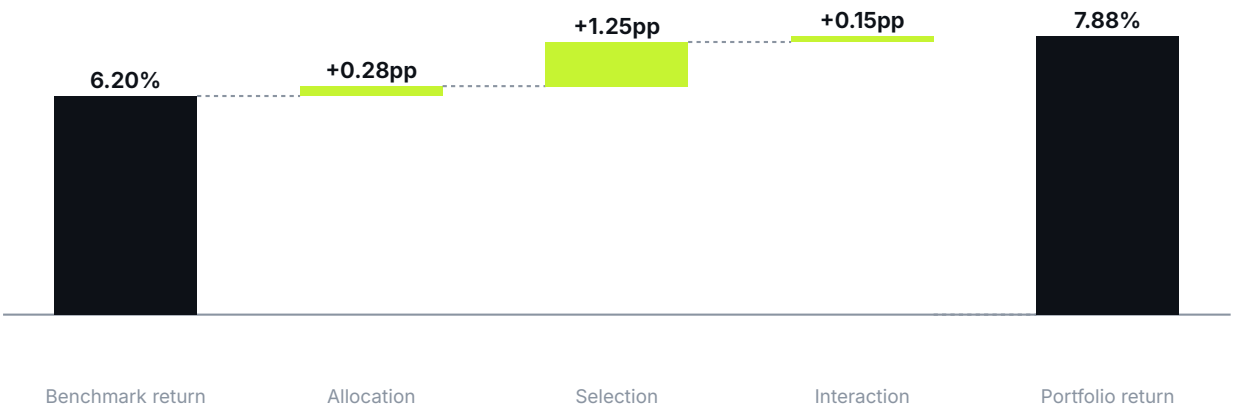
Returned evidence only; no local subtraction or accumulation.

REALIZED SUMMARY EFFECTS



Each plotted value is a returned summary field S2.

EXPECTED SUMMARY EFFECTS



Each plotted value is a returned summary field S3.

Key judgment — Historical

Observation: Selection is the dominant returned realized shortfall, driven by Alternatives S2. Why it matters: It bridges the +2.75% benchmark return to the -3.35% portfolio return. Uncertainty: One realized window may not persist. What would change the view: A later governed period with a different returned selection effect.

Key judgment — Ex ante

Observation: Selection is the dominant returned expected gain, driven by Alternatives S3. Why it matters: It bridges the 6.20% expected benchmark return to the 7.88% expected portfolio return. Uncertainty: Inputs are governed assumptions. What would change the view: Refreshed inputs with a different returned selection effect.

Key judgment — Comparison

Observation: Comparison capability is unavailable S1. Why it matters: Persistence cannot be assessed. Uncertainty: A future tool may return metrics. What would change the view: Published capability.

02 Scope and Compatibility

Portfolio, benchmark, cohort and currency identifiers match

Compatible for display-only fallback

Portfolio, benchmark role, parent cohort and calculation currency match S2 S3.

Compatibility gate	● **available**	Four identifiers match S2 S3

03 Historical Attribution

Returned realized effects are shown without recalculation

Historical effects by segment

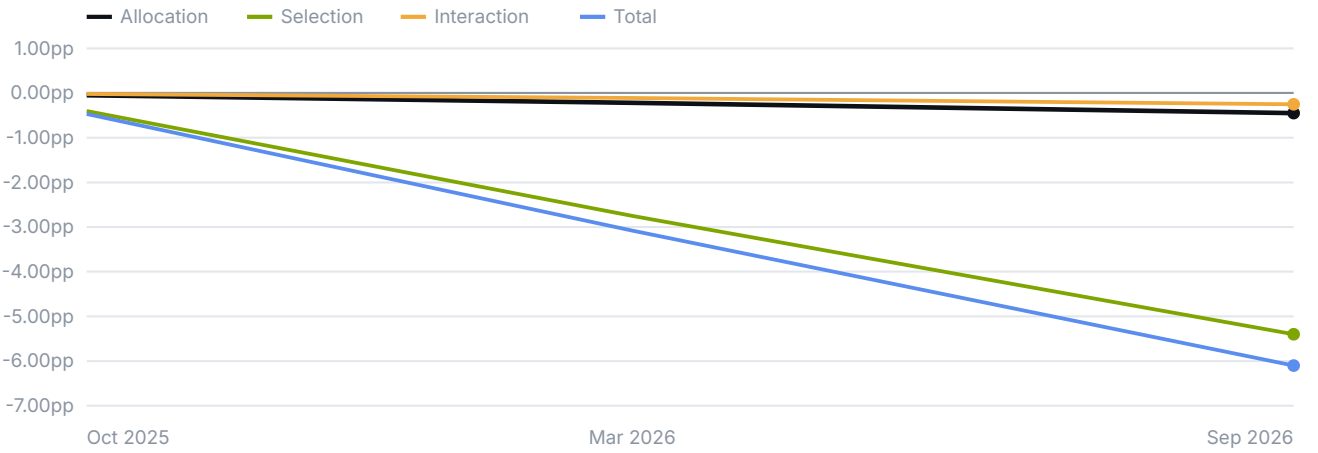
Not available — stacked values contain no positive total

HISTORICAL PORTFOLIO AND BENCHMARK AVERAGE WEIGHTS



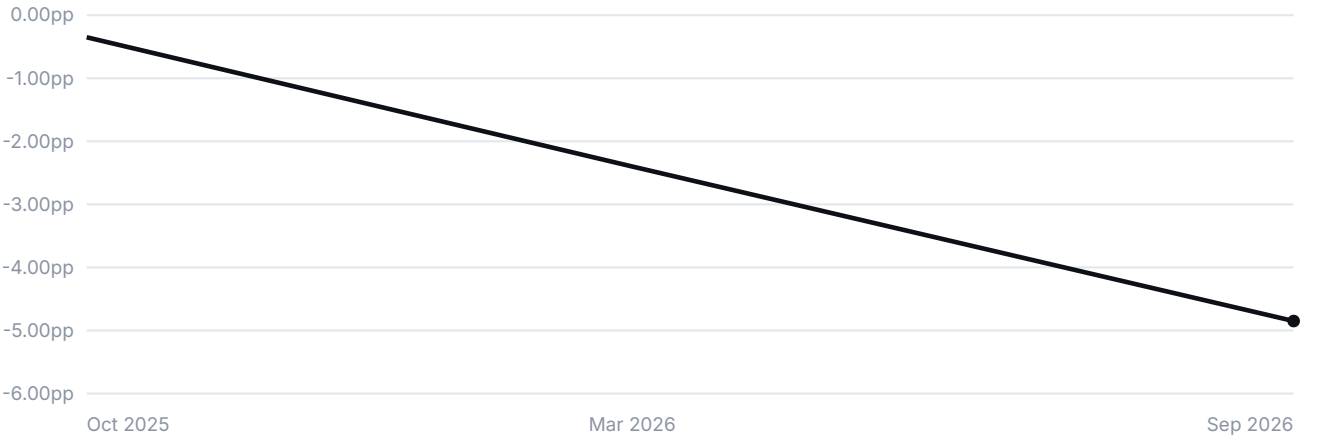
Returned arithmetic average beginning weights; no local difference S2.

CUMULATIVE LINKED EFFECTS



Server-returned cumulative linked fields; no local accumulation S2.

CUMULATIVE ALTERNATIVES SELECTION



Returned rows filtered to allocation_name=Alternatives; no local accumulation S2.

Returned cumulative growth context

Not available — cumulative_growth was not returned.

Analysis — Selection leads detractors

Observation: Returned selection is the -5.40pp dominant realized shortfall and Alternatives contributes -4.85pp S2. Why it matters: Selection drives the returned bridge from +2.75% benchmark return to -3.35% portfolio return. Uncertainty: The evidence covers one linked window. What would change the view: A later historical result with a different returned selection effect.

Benchmark and excess boundary

Returned diagnostics.entity_return_reconciliation keeps attribution-summary excess against the linked segment benchmark composite separate from historical-return excess against the policy benchmark entity S2.

04 Governed Ex Ante Attribution

Returned expected effects are shown as assumptions

EXPECTED EFFECTS BY SEGMENT



Returned expected allocation, selection and interaction by segment S3.

EX ANTE ASSUMPTIONS

assumption_set_name	**2026 Strategic CMA** S3
regime_name	**user-defined regime** S3
horizon	**1 year** S3
portfolio_return_source	**allocation_expected_return_base** S3
benchmark_return_source	**benchmark_expected_return_base** S3

EXPECTED PORTFOLIO AND BENCHMARK RETURNS



Returned expected returns grouped without local subtraction S3.

Analysis — Selection leads expectations

Observation: Returned selection is the +1.25pp dominant expected gain and Alternatives contributes +1.05pp S3. Why it matters: Selection drives the returned bridge from 6.20% expected benchmark return to 7.88% expected portfolio return. Uncertainty: Inputs are governed assumptions. What would change the view: A refreshed ex ante result with a different returned selection effect.

05 Simulated Attribution

Simulation capability is planned

Not available

Not available — planned per get_gradient_capabilities S1 .

Capability	**unavailable**	Planned per capability source S1
Invocation	**unavailable**	Not called because capability is unavailable S1

Analysis — Simulation unavailable

Observation: Simulation is planned per the capability source S1 . Why it matters: Scenario effects cannot be assessed. Uncertainty: Capability readiness may change. What would change the view: A capability response marking the exact simulation tool available.

06 Cross-Lane Comparison

Returned lanes are displayed while comparison remains planned

Not available

Not available — planned per get_gradient_capabilities S1 .

RETURNED CROSS-LANE EFFECT MATRIX

	REALIZED	EXPECTED
ALLOCATION	0.20	0.40
SELECTION	-0.80	-0.10
INTERACTION	0.00	0.05

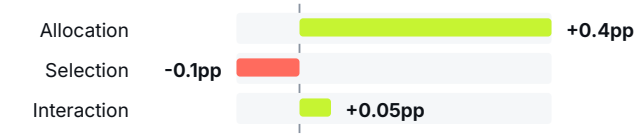
Side-by-side returned summary fields; no local differences S2 S3 .

REALIZED RETURNED EFFECTS



Displayed unchanged S2 .

EXPECTED RETURNED EFFECTS



Displayed unchanged S3 .

Unavailable comparison metrics

Differences, concentration, persistence, and Sources that persist are unavailable without a governed comparison result S1 .

Lane differences	**unavailable**	Requires governed comparison output S1
Concentration	**unavailable**	No local aggregation permitted S1
Persistence	**unavailable**	No server persistence metric S1
Sources that persist	**unavailable**	No governed comparison result S1

Analysis — Comparison unavailable

Observation: Comparison is planned per get_gradient_capabilities S1 . Why it matters: No persistence claim is supported. Uncertainty: Planned capability timing is not returned. What would change the view: A governed comparison result.

07 Coverage and Limitations

Two live lanes and two planned capabilities define coverage

Historical	**available**	Governed result S2
Ex ante	**available**	Governed result S3
Planned tools	**unavailable**	Simulation and comparison are planned S1

A Appendix — Sources and methods

Governed source inventory

TAG	EVIDENCE	BASIS	TOOL / VIEW	KEY PARAMETERS	AS OF	DATA SCOPE	VALIDATION	DIGEST
S1	Capability gate	gradient_tool	get_gradient_capabilities	illustrative sample	2026-09-30	Gradient-governed review evidence	passed	sha256:aaaaaaaaaaaa
S2	Historical attribution	gradient_tool	get_portfolio_attribution	illustrative sample	2026-09-30	Gradient-governed review evidence	passed	sha256:bbbbbbbbbbbb
S3	Ex ante attribution	gradient_tool	get_portfolio_ex_ante_attribution	illustrative sample	2026-09-30	Gradient-governed review evidence	passed	sha256:cccccccccccc

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

Full digests — S1: sha256:aaaaaaaaaaa